Moving to another question will save this response.

Question 1

Analyzing algorithms is no longer important because computers have gotten so fast.

- True
- False

Moving to another question will save this response.

Moving to another question will save this response.

Question

Question 2

0.5 poin

If $f(n) = 2^n + n$, then:

- $f(n) = O(2^n)$
- $f(n) = O(n^n)$
- $f(n) = O(2^n+n)$
- All the above answers are correct.

Moving to another question will save this response.

Question 2

Moving to another question will save this response.

Question 4

0.5 points

4- If $f(n) = 3^n$ and $g(n) = 3^{(n^2)}$, then

- f(n) = O(g(n))
- f(n) = Ω(g(n))
- f(n) = Θ(g(n))
- All the above answers are correct.

Moving to another question will save this response.

Question 4 of 10

⚠ Moving to another question will save this response.

Question 5

0.5 points

If $f(n) = 3^n$ and $g(n) = (\sqrt{n})^n$, then

- f(n) = O(g(n))
- f(n) = Ω(g(n))
- f(n) = Θ(g(n))
- All the above answers are correct.

⚠ Moving to another question will save this response.

Question Completion Status:

1 2 3 4 5 6 7 8 9 10

Moving to another question will save this response.

Quest

Question 6

0.5 p

The values $c = 2$ And $n_0 = 1$ can be used to prove that $f(n) = O(n^3)$ where $f(n) = n^3 + 1$.

- True
- False

Moving to another question will save this response.

Questio

Moving to another question will save this response.

Question

Question 7

0.5 poin

Which of the following statement is not correct:

- f(n) = 30n+8 is $O(n^2)$
- f(n) = 30n+8 is $\Omega(1)$
- f(n) = 30n +8 is $\omega(n)$
- f(n) = 30n+8 is $O(n)$

Moving to another question will save this response.

Question 7

Moving to another question will save this response.

Question

Question 8

0.5 po

If $f(n) = (2n^7 + n^4 + 10n^3)(3m^4 + 2m^3 + 1)$, then:

- $f(n) = O(n^{11})$
- $f(n) = O(n^7 m^4)$
- $f(n) = O(n^7)$
- All the above answers are incorrect.

```
            k = k +1
        }
        t = 0
        while (t< n)
        {
            t = t +2
        }
}
```

The time complexity of the previous algorithm is:

- (●) n log n
- ( ) n
- ( ) n^2
- ( ) ½ n + n

Save and Submit

Click **Submit** to complete this assessment.

Question **10** of **10**

Question 10

0.5 points Saved

Consider two different algorithms A and B. Both algorithms have been implemented using the same programming language. Using inputs with the same size, if the running time of A is 0.5 millisecond while the running time of B is 1 millisecond, then

- A is faster than B
- B is faster than A
- A and B have the same speed
- Cannot be determined

Click **Submit** to complete this assessment.

Question **10** of **10**